

INCREASING PRODUCTIVITY AND QUALITY With New-Age Soldering Irons

Thinking of how to speed up your assembly lines? These IoT based soldering irons with easy-to-use menu, stable temperature, and sleep mode features might help. This article discusses some innovative soldering trends that can elevate assembly to a new level

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Soldering is a critical part of the production chain or the assembly line that can completely change the fortune of a manufacturing organisation. With the new trends and innovations in the electronics industry, the traditional soldering stations in an assembly line need to be updated.

There are numerous innovations taking place in the arena of soldering. For example, the assembly lines can now be managed by just tracing the performance and precision of the soldering work. Not just that, there are special soldering tips being introduced for hard-to-reach areas in a chip or a system. These innovations, if utilised well, can completely change the pace of production for good.

Many soldering iron manufacturers are now designing their irons with ergonomics in mind. Ergonomic soldering irons are designed to fit comfortably in your hand, reducing the risk of strain or injury. They often feature comfortable grips, adjustable stands, and other accessories that make them easier and safer to use. Shorter grip-to-tip distance allows more control.

Let us have a look at some of the innovations, so you can make an informed decision when purchasing your soldering equipment.

Heat control

Heat control manages temperature across various components as higher temperatures are needed for soldering larger or thicker components while lower temperatures are needed for more delicate work or for working with heat-sensitive components.

The 'Quick Heat' feature in PX-280 soldering iron from Goot reaches temperature in time comparable to that of middle-class soldering stations. The soldering temperature ranges from 200 to 500°C. It reaches 350°C in just 30 seconds with a standard tip. With a high-power heater and sensor, there is almost no temperature fluctuation during heavy-duty continuous operation.

The WE 1010 soldering iron from Weller Tools offers excellent temperature stability. It features a temperature lock that stabilises heat levels and prevents them from wavering. The device can operate at between 100

and 450°C. It keeps temperatures at plus or minus six degrees Celsius of the intended temperature.

While most soldering irons start at 100°C, the B-Iron from JBC Tools offers a minimum of 30°C, thereby reducing the risk of damaging



Fig. 1: A few shapes of soldering tips (Credit: JBC Tools)